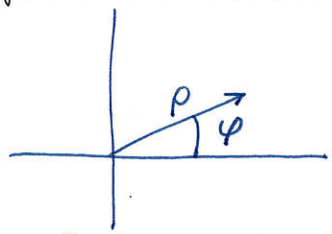
Cylindrical coordinates

$$\Delta V = -\frac{\rho}{\varepsilon_0}$$



$$\underbrace{\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial V}{\partial\rho}\right)}_{\frac{1}{\rho}\left(\frac{\partial V}{\partial\rho}+\rho\frac{\partial^2 V}{\partial\rho^2}\right)} + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial\varphi^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

$\boxed{2d} \Rightarrow \vec{r} \Rightarrow (\rho, \varphi)$: $V(\rho,\varphi) = R(\rho)\,\Phi(\varphi)$

$$\frac{\Phi(\varphi)}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial R}{\partial\rho}\right) + \frac{R(\rho)}{\rho^2}\frac{\partial^2\Phi}{\partial\varphi^2} = 0$$

$$\underbrace{\frac{\rho}{R}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial R}{\partial\rho}\right)}_{\nu^2} + \underbrace{\frac{1}{\Phi}\frac{d^2\Phi}{d\varphi^2}}_{-\nu^2} = 0$$

$$\frac{d^2\Phi}{d\varphi^2} = -\nu^2\Phi$$

$$\Phi(\varphi) = a_0 + b_0\varphi + \sum_\nu\left[a_\nu\cos(\nu\varphi) + b_\nu\sin(\nu\varphi)\right]$$

radial: $\rho\frac{\partial}{\partial\rho}\left(\rho\frac{\partial R}{\partial\rho}\right) = \nu^2 R$

$$\rho^2\frac{\partial^2 R}{\partial\rho^2} + \rho\frac{\partial R}{\partial\rho} - \nu^2 R = 0 \quad \text{(leads to Bessel's)}$$

Try $R = \alpha_\nu\rho^\nu + \beta_\nu\rho^{-\nu}$

$$\frac{\partial}{\partial \rho}\left[\rho\left(\nu \alpha_\nu \rho^{\nu-1} - \nu \beta_\nu \rho^{-\nu-1}\right)\right] = \nu^2 \frac{\alpha_\nu \rho^\nu + \beta_\nu \rho^{-\nu}}{\rho}$$

$$\underbrace{\frac{\partial}{\partial \rho}\left(\nu\left(\alpha_\nu \rho^\nu - \beta_\nu \rho^{-\nu}\right)\right)}_{\nu^2 \alpha_\nu \rho^{\nu-1} + \nu^2 \beta_\nu \rho^{-\nu-1}} = \nu^2\left(\alpha_\nu \rho^{\nu-1} + \beta_\nu \rho^{-\nu-1}\right)$$

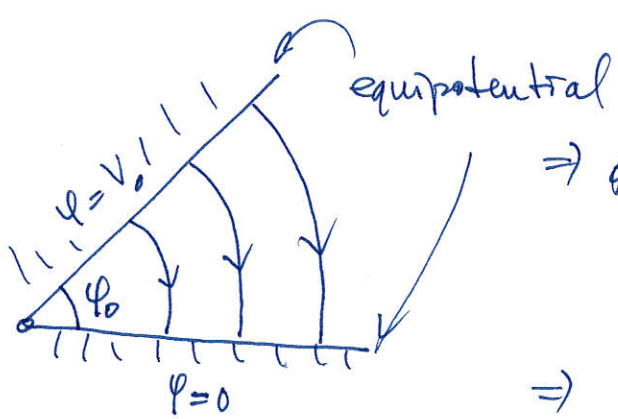
$$\Rightarrow V(\rho, \varphi) = (a_0 + b_0 \varphi) R_0(\rho)$$

$$+ \sum_\nu \left(\alpha_\nu \rho^\nu + \beta_\nu \rho^{-\nu}\right)\left[a_\nu \cos(\nu\varphi) + b_\nu \sin(\nu\varphi)\right]$$

If $\nu = 0$: $\quad \frac{\partial}{\partial \rho}\left(\rho \frac{\partial R}{\partial \rho}\right) = 0$

$$\frac{\partial R}{\partial \rho} = \frac{c}{\rho} \Rightarrow R = \alpha_0 + \beta_0 \ln \rho$$

$$R' = \frac{\beta_0}{\rho}$$



$$\Rightarrow V(\rho, \varphi = 0) = 0$$

$$V(\rho, \varphi = \varphi_0) = V_0$$

$$\Rightarrow \alpha_\nu = \beta_\nu = 0.$$

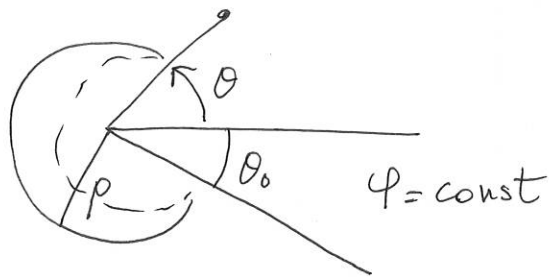
$$V(\rho, \varphi) = (a_0 + b_0 \varphi)\alpha_0$$

$$V(\rho, \varphi = 0) = 0 \Rightarrow a_0 = 0$$

$$V(\rho, \varphi) = \frac{V_0 \varphi}{\varphi_0} \Rightarrow E = -\frac{1}{\rho}\frac{V_0}{\varphi_0}$$

singularity is put off at ρ_0.

Conducting wedge



$$V(\rho,\theta) = (a_0 + b_0\,\theta)\,k_0 + \sum_\nu \left(\alpha_\nu \rho^\nu + \beta_\nu \rho^{-\nu}\right) \cdot$$
$$\cdot \left[a_\nu \cos(\nu\theta) + b_\nu \sin(\nu\theta)\right]$$

BC: $V(\rho, \theta=0) = V(\rho, \theta_1 = 2\pi - \theta_0)$

$\Rightarrow a_0 = b_0 = 0$, $\beta_\nu = 0$ (must be regular for $\rho \to 0$)

$$V(\rho,\theta) = \sum_\nu \alpha_\nu \rho^\nu \left[a_\nu \cos(\nu\theta) + b_\nu \sin(\nu\theta)\right]$$

$$V(\rho,0) = \sum_\nu \alpha_\nu \rho^\nu \left[a_\nu + \cancel{b_\nu \cdot 0}\right]$$

$$V(\rho,\theta_1) = \sum_\nu \alpha_\nu \rho^\nu \left[a_\nu \cos(\nu\theta_1) + b_\nu \sin(\nu\theta_1)\right]$$

$\Rightarrow \cos(\nu\theta_1) = 1$, $\nu_k \theta_1 = 2\pi k$

$$\nu_k = \frac{2\pi k}{\theta_1}$$

$\Rightarrow \sin(\nu\theta_1) = 0$, $\nu_\ell \theta_1 = \pi \ell$

$$\nu_\ell = \frac{\pi \ell}{\theta_1}$$

$$\Rightarrow V(\rho,\theta) = \sum_\nu \alpha_\nu \rho^\nu \left[a_\nu \cos\left(2\pi k \frac{\theta}{\theta_1}\right) + b_\nu \sin\left(\pi \ell \frac{\theta}{\theta_1}\right)\right]$$

$$\nu = \frac{2\pi k}{\theta_1} \qquad \nu = \frac{\pi \ell}{\theta_1}$$

Electric field $\vec{E} = -\vec{\nabla} V = -\left\{ \frac{\partial V}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial V}{\partial \theta} \hat{\theta} \right\}$

$$= -\sum_{\nu} \alpha_\nu \nu \rho^{\nu-1} \left[a_\nu \cos\left(\frac{\nu_k \theta}{\theta_1}\right) + b_\nu \sin\left(\frac{\nu_\ell \theta}{\theta_1}\right) \right] \hat{\rho}$$

$$+ \alpha_\nu \rho^{\nu-1} \left[-\frac{\nu_k}{\theta_1} a_\nu \sin\left(\frac{\nu_k \theta}{\theta_1}\right) + b_\nu \frac{\nu_\ell}{\theta_1} \cos \frac{\nu_\ell \theta}{\theta_1} \right] \hat{\theta}.$$

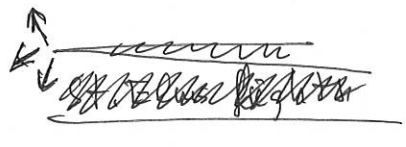
$\sim \rho^{\nu-1} \Rightarrow$ strongest divergence comes from smallest $\nu = \nu_{\ell=1} = \frac{\pi}{\theta_1}$

$$E \sim (\rho)^{\frac{\theta_0 - \pi}{2\pi - \theta_0}} = \rho^{\nu-1}$$

$$\nu - 1 = \frac{\pi}{\theta_1} - 1 = \frac{\pi}{2\pi - \theta_0} - 1 = \frac{\pi - 2\pi + \theta_0}{2\pi - \theta_0} = \frac{\theta_0 - \pi}{2\pi - \theta_0}$$

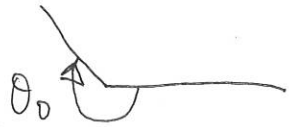
(1) Now, $\nu - 1 < 0$ (diverging field as $\rho \to 0$) for $\theta_0 < \pi$.

In particular, very thin wedge $\theta_0 \to 0$ $\nu - 1 \to \frac{-\pi}{2\pi} = -\frac{1}{2}$

$E \to \frac{1}{\sqrt{\rho}}$  St. Elmo's fire

(2) $\theta_0 = \pi \Rightarrow \nu - 1 = 0 \Rightarrow E \sim$ const.

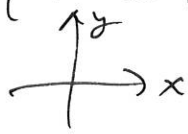
(3) $\theta_0 > \pi \Rightarrow \nu - 1 > 0 \Rightarrow E \sim \rho^{\frac{\theta_0 - \pi}{2\pi - \theta_0}} \xrightarrow[\rho \to 0]{} 0$



Conformal mapping: $\Phi(\vec{r}) = \Phi(x,y)$ only.

<u>Complex analysis</u> (works for "2-d" electrostatics problems)

$z = x + iy$



when Φ is z-independent $\Rightarrow$ charges are extended along z axis as wires, cylinders, etc.

complex variable function $F(z)$. E.g. $F(z) = z$

$F(z) = z^2$ or $\frac{1}{z}$.

$$F(z) = F(z = x+iy) = \underset{\uparrow}{U(x,y)} + i\,\underset{\uparrow}{V(x,y)}$$

real functions

e.g. for $F = z^2 = (x+iy)^2 = \underbrace{x^2 - y^2}_{U} + i\,\underbrace{2xy}_{V}$

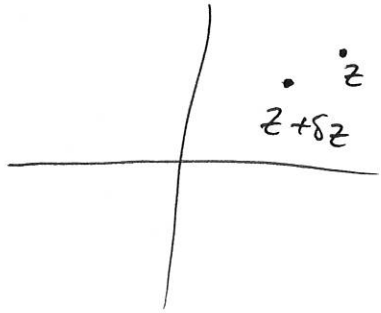
How do we introduce derivative?

$$\frac{dF}{dz} = \lim_{|\delta z| \to 0} \frac{F(z+\delta z) - F(z)}{\delta z}$$

↳ which direction on the z-plane to choose.

For (well-behaved) ~~analytic~~ holomorphic functions

trajectory of δz does not matter: the limit (derivative) does not depend on direction taken for δz.



But this implies some restrictions on U and V!

Let $\delta z = \delta x$ ($y = \text{const}$) $\Rightarrow$

$$\frac{dF}{dz} = \lim_{\delta x \to 0} \frac{U(x+\delta x, y) + iV(x+\delta x, y) - U(x,y) - iV(x,y)}{\delta x} = \frac{\partial U}{\partial x} + i\,\partial V/\partial x$$

Now let $\delta z = i\delta y$ ($x = \text{const}$) $\Rightarrow$

$$\frac{dF}{dz} = -i\left(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right) = -i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

$\Rightarrow$ <u>must be</u> $$\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = -i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}$$

$$\Rightarrow \begin{cases} \dfrac{\partial u}{\partial x} = \dfrac{\partial v}{\partial y} \\[2ex] \dfrac{\partial v}{\partial x} = -\dfrac{\partial u}{\partial y} \end{cases}$$

Cauchy-Riemann relations

But now $$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} = \frac{\partial}{\partial y}\left(-\frac{\partial u}{\partial y}\right) = -\frac{\partial^2 u}{\partial y^2}$$

$$\frac{\partial^2 v}{\partial x^2} = -\frac{\partial^2 u}{\partial x \partial y} = -\frac{\partial}{\partial y}\frac{\partial v}{\partial y} = -\frac{\partial^2 v}{\partial y^2}$$

$$\Rightarrow \begin{cases} \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0 \\[2ex] \dfrac{\partial^2 v}{\partial y^2} + \dfrac{\partial^2 v}{\partial y^2} = 0 \end{cases}$$

Harmonic functions
$\downarrow$
such functions automatically satisfy Laplace eq. in $d=2$.

$\Rightarrow$ $u(x,y)$ and $v(x,y)$ describe some <u>possible</u> electric potentials

Moreover $$\vec{\nabla} u \cdot \vec{\nabla} v = \frac{\partial u}{\partial x}\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x}\left(-\frac{\partial u}{\partial y}\right) + \frac{\partial u}{\partial y}\left(\frac{\partial u}{\partial x}\right) = 0$$

$\Rightarrow$ contours of u are perpendicular to those of v.

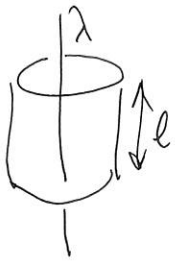
Ex: logarithmic function

$$f(z) = A \ln z = A(\ln r + i\theta)$$

$$z = x + iy = re^{i\theta}$$

$$\Rightarrow \quad u(x,y) = A \ln r = \frac{A}{2} \ln(x^2+y^2) \leftarrow \text{potential of a charged wire}, \; A = -\frac{\lambda}{2\pi\varepsilon_0}$$

$$v(x,y) = A\theta = A \arctan\left(\frac{y}{x}\right)$$



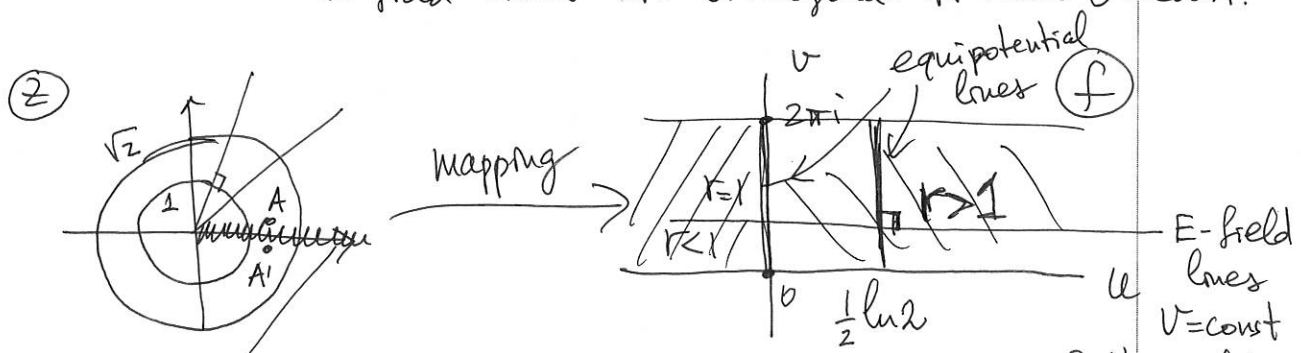
$$E(r)\, 2\pi r\, l = \frac{\lambda l}{\varepsilon_0}$$

$$E(r) = \frac{\lambda}{2\pi\varepsilon_0 r}$$

$$\Rightarrow \varphi(r) = -\frac{\lambda}{2\pi\varepsilon_0} \ln\left(\frac{r}{r_0}\right)$$

$\Rightarrow u = \varphi$: equipotential lines are circles $r = \text{const}$

E-field lines are orthogonal to them $\theta = \text{const}$.



Orthogonality of $\varphi = \text{const}$, $\vec{\nabla}\varphi = \text{const}$ is preserved.

Important: $f(z)$ is not analytic at $z=0$ where its derivative diverges. $\Rightarrow v$ is multivalued

Introduce branch-cut which is not allowed to be crossed.

$\Rightarrow$ cut along $x \in [0,\infty]$.

$v \to v + 2\pi A$ when θ changes by 2π by going around $r=0$ (singular point)

The cut makes ln-function to be single-valued: $0 \le \theta \le 2\pi$

$\theta_A = 0$ while $\theta_{A'} = 2\pi$

Unit circle $r=1$ $\&$, $\theta=(0,2\pi)$, maps onto a segment $0 \le v = \mathrm{Im}(f) \le 2\pi i$

It is usually simpler to figure out what arrangements of conductors corresponds to a given analytic function $f(z)$ then to solve the inverse problem.

$\Rightarrow$ ~~U can describe el. potential and V shows direction of corresponding electric field.~~

So, consider finally $F(z) = -E_0 z + \frac{E_0 c^2}{z}$

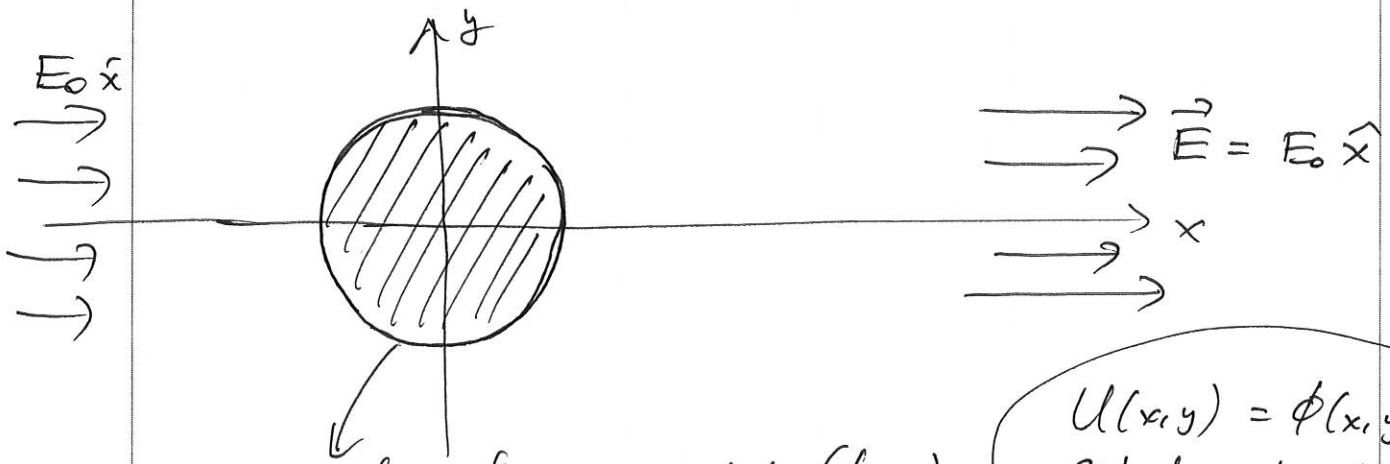
$$= -E_0 (x+iy) + \frac{E_0 c^2 (x-iy)}{x^2+y^2}$$

$$\Rightarrow U(x,y) = -E_0 x + \frac{E_0 c^2 x}{x^2+y^2}$$

$$V(x,y) = -E_0 y - \frac{E_0 c^2 y}{x^2+y^2}$$

For $x \to \infty$ $U(x,y) = -E_0 x$ describes ϕ of uniform el. field : $\vec{E} = -\vec{\nabla}\cdot\phi \cong -\vec{\nabla}(-E_0 x) = E_0 \hat{x}$

$U(x,y) = 0$ for $x^2+y^2 = c^2$ (and also $x=0$).



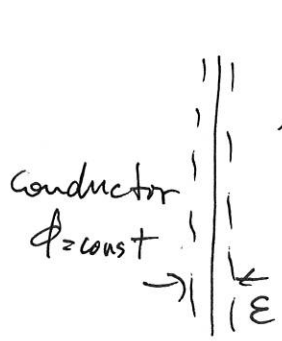
surface of a grounded ($\phi = 0$) conducting cylinder!

$U(x,y) = \phi(x,y)$ satisfies Laplace eq. for this problem everywhere for $x^2+y^2 \geq c^2$.

$$\Rightarrow \phi(x,y) = -E_0 x + \frac{E_0 c^2 x}{x^2+y^2}$$ describes grounded conducting cylinder in uniform el. field

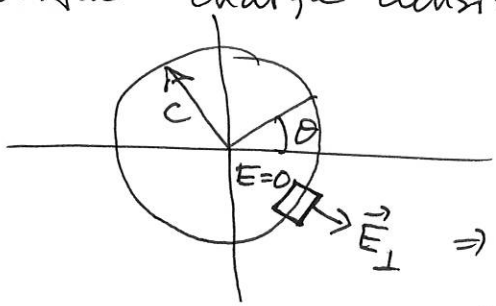
Surface charge density from Poisson's eq:

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = -\frac{\rho}{\varepsilon_0}$$



$$\lim_{\varepsilon\to 0}\int_{-\varepsilon}^{\varepsilon} dx\,(\Delta\phi) = \frac{d\phi}{dx}\Big|_{-0}^{+0} = -\frac{1}{\varepsilon_0}\underbrace{\int dx\,\rho}_{\sigma}$$

Induced Surface charge density



on the surface

$$\Rightarrow E_\perp = \frac{\sigma}{\varepsilon_0}$$

Note that $\vec{E} = \vec{E}_\perp$ but $E_\perp = E_\perp(\theta)$.

$$E_\perp = E_r(r=c,\theta)$$

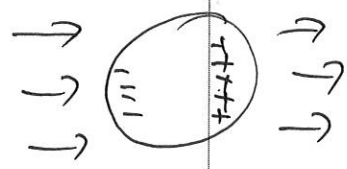
$$\vec{E} = -\vec{\nabla}\cdot\phi = -\left(\frac{\partial\phi}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial\phi}{\partial\theta}\hat{\theta} + \cancel{\frac{\partial\phi}{\partial z}\hat{z}}\right)$$

$$\Rightarrow E_r = -\frac{\partial\phi}{\partial r}$$

But $\phi = -E_0 x + \dfrac{E_0 c^2 x}{x^2+y^2} = -E_0\cos\theta\, r + \dfrac{E_0 c^2\cos\theta}{r}$

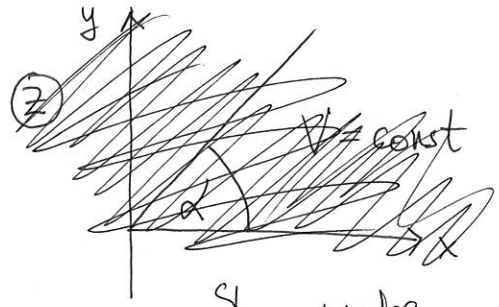
$$x = r\cos\theta$$

$$-E_r = +\frac{\partial\phi}{\partial r} = -E_0\cos\theta - \frac{E_0 c^2\cos\theta}{r^2}$$

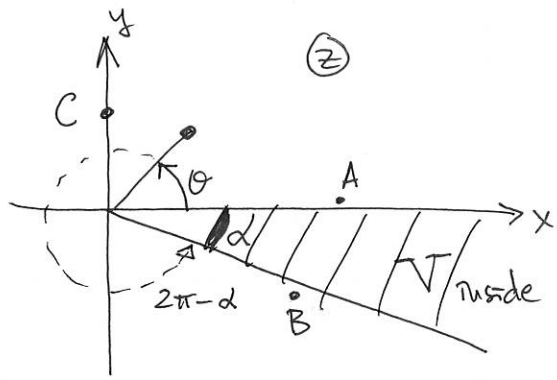


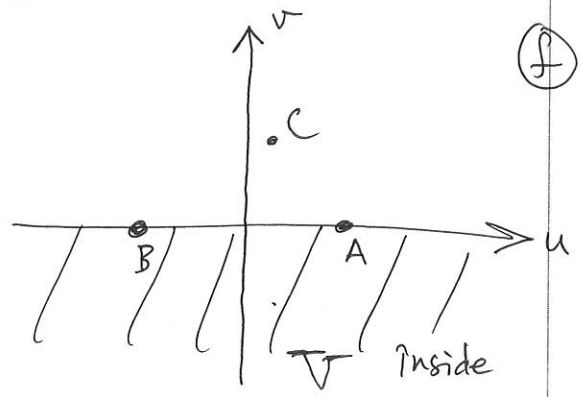
$$\Rightarrow \sigma = \varepsilon_0 E_r(r=c) = \varepsilon_0\left(E_0\cos\theta + \frac{E_0 c^2\cos\theta}{c^2}\right) = 2\varepsilon_0 E_0\cos\theta$$

Total charge $\rho = \int\sigma\, d\theta = 0$



Sharp wedge





Consider $f(z) = z^{\frac{\pi}{2\pi-\alpha}}$

$$z = re^{i\theta} \Rightarrow f(z) = r^{\frac{\pi}{2\pi-\alpha}} e^{i\frac{\theta\pi}{2\pi-\alpha}}$$

A: $\theta = 0,\ r_A \longrightarrow r_A^{\frac{\pi}{2\pi-\alpha}} = f \Rightarrow u = r_A^{\frac{\pi}{2\pi-\alpha}},\ v = 0.$

B: $\theta = 2\pi-\alpha,\ r_B \rightarrow f = r_B^{\frac{\pi}{2\pi-\alpha}} e^{i\frac{(2\pi-\alpha)\pi}{2\pi-\alpha}} = r_B^{\frac{\pi}{2\pi-\alpha}} e^{i\pi} =$

$= -r_B^{\frac{\pi}{2\pi-\alpha}} = u,\ v = 0$

C: $\theta = \frac{\pi}{2},\ r_c \rightarrow f = (r_c)^{\frac{\pi}{2\pi-\alpha}} e^{i\frac{\pi^2}{2(2\pi-\alpha)}}$

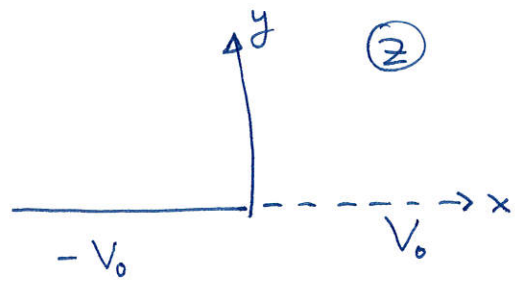
$$\frac{\pi}{4} < \frac{\pi^2}{2(2\pi-\alpha)} < \frac{\pi}{2} \quad \text{for } 0 < \alpha < \pi$$

$\Rightarrow \sin\frac{\pi^2}{2(2\pi-\alpha)} > 0 \Rightarrow$ part C maps onto $v > 0$ part of the f-plane

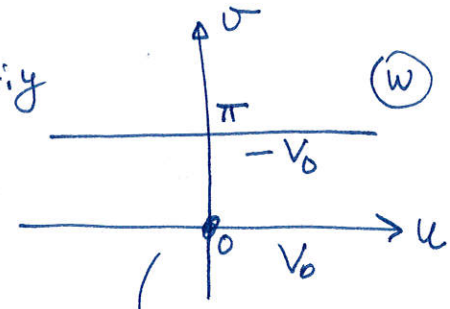
$$\Rightarrow u = \mathrm{Re}(f(z)) = r^{\frac{\pi}{2\pi-\alpha}} \cos\left(\frac{\pi\theta}{2\pi-\alpha}\right) = r^{\frac{1}{2-\alpha/\pi}} \cos\left(\frac{\theta}{2-\frac{\alpha}{\pi}}\right)$$

describes potential outside the wedge.

$$E_r = -\frac{d\varphi}{dr} \sim r^{\frac{1}{2-\frac{\alpha}{\pi}}-1} \cos\left(\frac{\theta}{2-\alpha/\pi}\right) \longrightarrow r^{-1/2} \text{ for } \alpha \to 0.$$



$z = x+iy$



$V(w) = V_0 - \frac{2V_0}{\pi} v$

$u+iv = w = \ln z \Rightarrow$

$y=0,\ x>0 \Rightarrow u+iv = \ln x \Rightarrow$ line $v=0$

$y=0,\ x<0 \Rightarrow u+iv = \ln\left(e^{i\pi}|x|\right) = \ln|x| + i\pi \Rightarrow$ line $v=\pi$

$$\ln(x+iy) = \ln\left(\rho e^{i\varphi}\right) = \ln\rho + i\varphi = u+iv$$

$$\Rightarrow v = \varphi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\tan\varphi = \frac{y}{x}$$

$$\Rightarrow V(x,y) = V_0 - \frac{2V_0}{\pi}\tan^{-1}\left(\frac{y}{x}\right) =$$

$$= \frac{2V_0}{\pi}\left(\frac{\pi}{2} - \tan^{-1}\left(\frac{y}{x}\right)\right) =$$

$$= \frac{2V_0}{\pi}\tan^{-1}\left(\frac{x}{y}\right)$$

$$V(x,y) = \frac{2V_0}{\pi}\begin{cases} \tan^{-1}(+\infty) \ (\text{for } x>0,\ y=0) = \frac{\pi}{2} \\ \tan^{-1}(-\infty) \ (x<0,\ y=0) = -\frac{\pi}{2} \end{cases}$$

satisfies original BC!